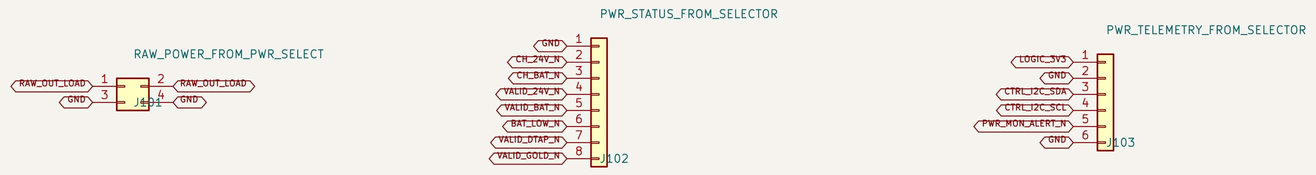
A. PWR-SELECT INTERFACE



Mates with PWR-SELECT J301; two contacts per polarity; validate 15 A derating.

Mates with PWR-SELECT J401; add 3.3 V pull-ups on the carrier.

Mates with PWR-SELECT J402; carrier supplies 3.3 V and owns I2C pull-ups.

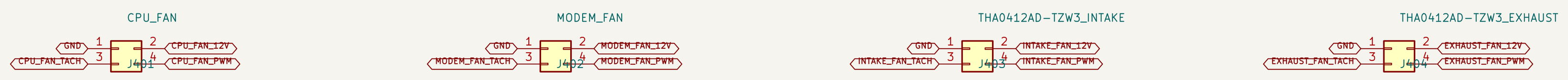
B. BUFFERED DIFFERENTIAL TDM + AUDIO CONTROL

Separate filtered audio-power harness; no fan/radio loads.

Mates with AUDIO-8X8 J101 using Molex 87832-6423 headers; harness assembly remains vibration-qualified.

Carrier drives MCLK/BCLK/FSYNC/DAC data; AUDIO-8X8 drives ADC data back.

C. FOUR INDEPENDENT PWM / TACH FANS



CPU_FAN

MODEM_FAN

THA0412AD-TZW3_INTAKE

THA0412AD-TZW3_EXHAUST

J403 is physically keyed/labeled INTAKE; J404 is keyed/labeled EXHAUST.

D. DETAILED CAPTURE SUITE – REV A1

01 CM5–Core–Allocated: exact 300–contact CM5 symbol, 76 owned contacts (74 connected, 2 assigned NC)

02 CM5–Core–Allocated: VCC_SYSIN, reset, recovery, power–on and debug UART

03 Network–PCIe: PI7C9X2G608GP, 3x LAN7430 and native WAN1 front end

04 Network–PCIe: 4x Wurth MagJack and Molex 0679101002 Mini PCIe 4T4R Wi–Fi

05 WWAN–SIM: TE 2199230–3 B–key, USB3/USB2, controls and dual–SIM mux

06 Display–Harness: HDMI, USB touch and dedicated 12 V / 2.5 A branch

07 Audio–Control: I2S0 TDM, I2S1 ES8316 headset and CTIA amplifier path

08 Thermal–IO: I2C translation, GPIO expansion, sensors and four fan channels

09 Power–Regulators–A1: protected raw hold–up, all main rails, sequencing and test points

10 Audio–8x8: all XLR shields bonded to chassis with one controlled AGND bond

11 Review: all three board roots have zero ERC violations; child context is allowlisted

DC/DC compensation and magnetics remain a separate power–design calculation milestone.

High–speed routing starts only after stackup and connector Z datums are released.

The two enclosure fans require an underside baffle to prevent direct intake/exhaust short flow.

E. CONNECTED CM5–CARRIER HIERARCHY

CM5 Core / Allocated Pins
File: CM5–Core–Allocated.kicad_sch

Network / PCIe / Wi–Fi
File: Network–PCIe.kicad_sch

WWAN / Dual SIM
File: WWAN–SIM.kicad_sch

Display Harness
File: Display–Harness.kicad_sch

Audio Control / Headset
File: Audio–Control.kicad_sch

Power Regulators
File: Power–Regulators–A1.kicad_sch

Thermal / Fans / IO
File: Thermal–IO.kicad_sch

J101/J201/J401–J404 are interface–only duplicates; the detailed–page connectors own the board footprints.

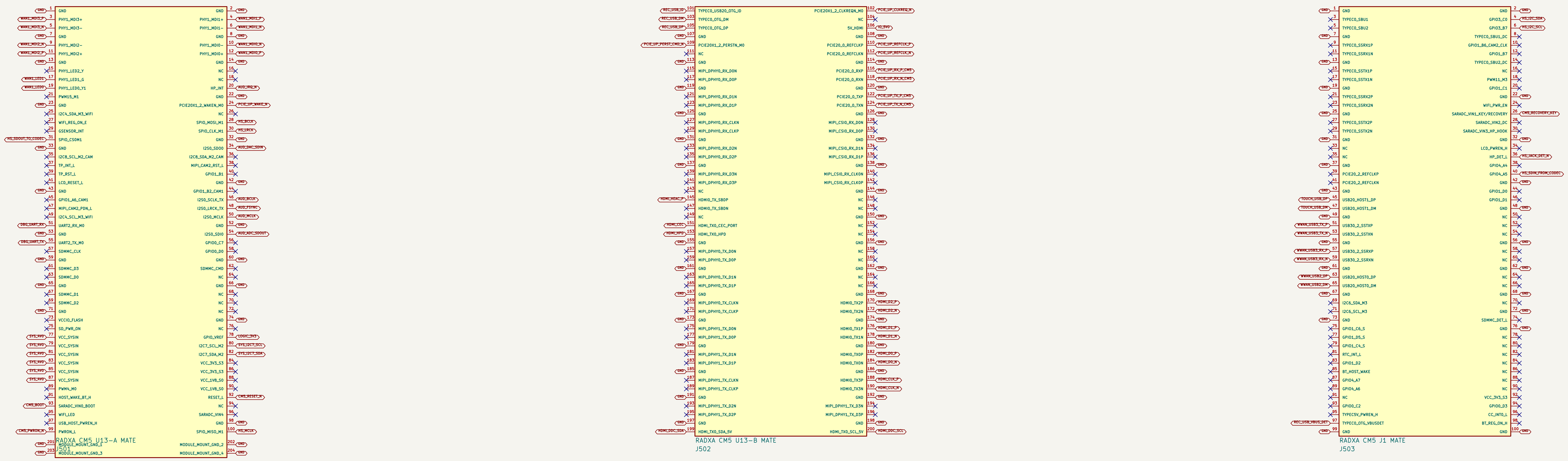
Global labels are intentional inter–page nets; prototype qualification gates remain
Program audio crosses a buffered differential TDM link
Fan 1 intake, fan 2 exhaust; independent PWM/tach channels
Physical CM5–CARRIER root; nested detailed pages form one connected board netlist
ProComm

Sheet: /
File: CM5–Carrier.kicad_sch

Title: Radxa CM5 ProComm Carrier – Interface Contract

Size: A1	Date: 2026–08–14	**Rev: A1**
KiCad E.D.A. 10.0.5–rc1		Id: 1/8

1. EXACT RADXA CM5 MATING CONNECTORS

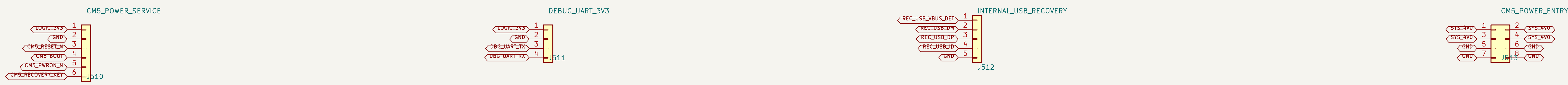


U13-A: 100 contacts + official grounded module pads 201–204

U13-B: 100 physical contacts

J1: 100 physical contacts

2. INTERNAL POWER / STARTUP / RECOVERY SERVICE



3.3 V UART only; keyed factory/debug cable.

Internal keyed recovery harness; service fixture owns the protected USB-C receptacle and CC network.

Kelvin-sense SYS_4V0 at J501; 4.0 V follows the Radxa CM5 carrier design note.

PCB lock: J501 and J502 share the U33 origin; J503 is +11.405, −25.415 mm at −90 degrees.

Internal keyed service header; never exposed on the top panel.

3. OWNERSHIP AUDIT

76 / 76 contacts owned; 74 connected, 2 assigned NC, 0 duplicate physical-pin claims.

WAN1 uses native MDI. PCIe Gen2 x1 feeds the packet switch.

WWAN owns USB30_2 + USB20_HOST0; touch owns USB20_HOST1.

HDMI_TX0, I2S0 TDM, I2S1 headset, I2C7, I2C3 and UART2 are reserved exactly once.

Pin 145 reaches HDMI pin 14; pin 147 is assigned NC because HEAC/ARC is unused. U13-B pin 106 receives IO_5V0.

Unused alternate-function contacts carry explicit no-connect markers on this A1 baseline.

DETAILED CAPTURE BASELINE — verify CM5 voltage limits before release
U33-A includes the four official grounded CM5 mounting pads; allocated GPIO domain is 3.3 V
76 contacts are owned: 74 connected and 2 assigned no-connect; all other functions are no-connect
Three DF40C-100DS mates; pad geometry and relative placement come from Radxa IO V2.20

ProComm

Sheet: /CM5 Core / Allocated Pins/
File: CM5-Core-Allocated.kicad_sch

Title: Radxa CM5 ProComm Carrier — Exact CM5 Connector Allocation

Size: A2	Date: 2026-08-13	**Rev: A1**
KiCad E.D.A. 10.0.5-rc1		Id: 2/8

Port 5 is disabled and unrouted; generic cable headers are prohibited on PCIe Gen2 lanes
Exact package balls/pins from Diodes DS40210 Rev 8 and Microchip DS00002631G
Native WAN1 plus three LAN7430 endpoints use exact Wurth 74991114412 MagJacks
PI7C9X2G608GP in 606 mode: x1 upstream plus five x1 downstream ports

ProComm

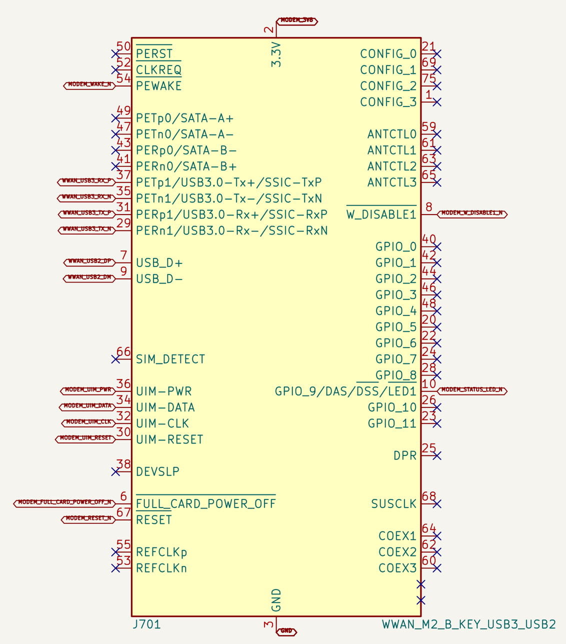
Sheet: /Network / PCIe / Wi-Fi/
File: Network-PCIe.kicad_sch

Title: Radxa CM5 ProComm Carrier — PCIe / Ethernet / Wi-Fi

Size: A1 | Date: 2026-08-14 | **Rev: A1**

KiCad E.D.A. 10.0.5-rc1 | Id: 3/8

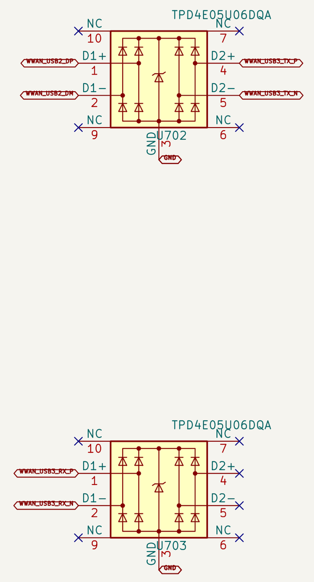
1. UNIVERSAL M.2 B-KEY WWAN SOCKET



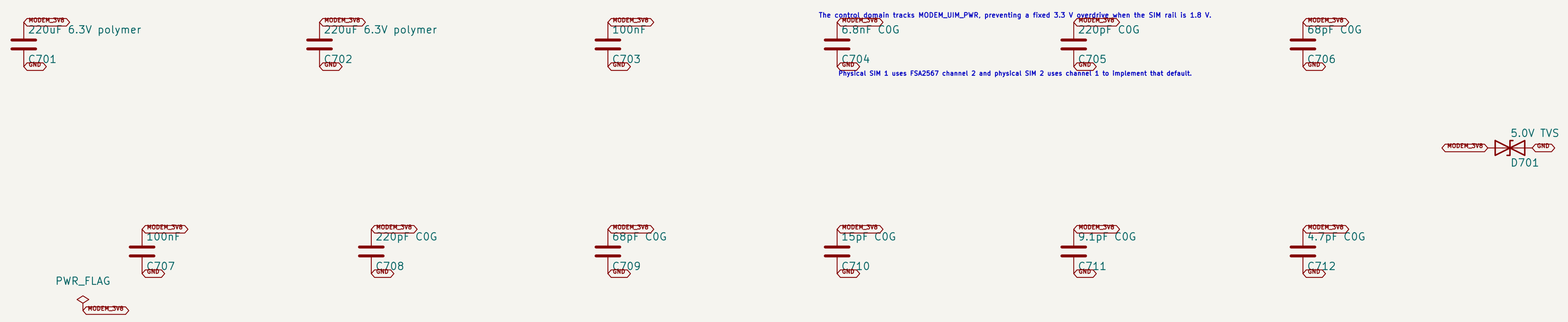
Socket power pins retain the stock symbol's 3.3 V names but are intentionally fed from MODEM_3V8.

Qualification window: 3.135–4.4 V module input, >=5 A transient capability, keyed B socket, 42 mm standoff.

2. USB ESD AND MODEM BULK CAPACITANCE



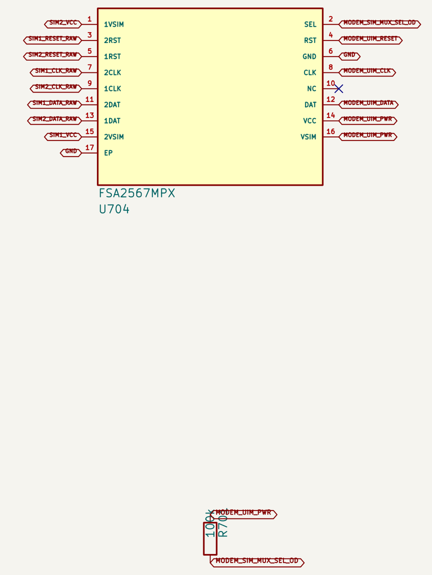
Place USB ESD at the modem socket; route USB 3 pairs as short, continuous 90 ohm differential channels.



C701–C712 and D701 implement the RM520N-GL two-bank VCC reference network directly at M.2 pins 2/4 and 70/72/74.

Power-Regulators C1189–C1192 retain 1320 uF upstream bulk; they do not replace the socket-local 2 x 220 uF and HF ladder.

3. FSA2567 DUAL-SIM MUX

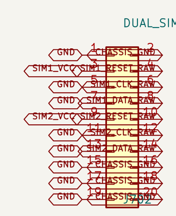


R701 keeps SEL high: SIM 1 is selected by default. GPIO control must be open drain; low selects SIM 2.

The control domain tracks MODEM_UIM_PWR, preventing a fixed 3.3 V overdrive when the SIM rail is 1.8 V.

Physical SIM 1 uses FSA2567 channel 2 and physical SIM 2 uses channel 1 to implement that default.

4. DIRECT DUAL-SIM DAUGHTERBOARD SOCKET



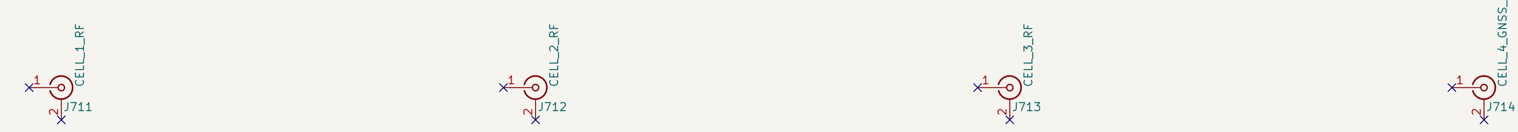
J702 plugs directly into SIM-SERVICE J1. There is no SIM cable harness; the horizontal daughterboard is fixed by four precision supports.

The 2.5 mm DF40 stack carries signals only. Four interleaved returns plus six auxiliary ground/chassis pins control loop area and shield impedance.

PWR_FLAG

PWR_FLAG

5. RF BULKHEAD HARNESS CONTRACT



CELL_1 CELL_2 CELL_3 CELL_4_GNSS

Four off-board ECT 818033349 USS RF IV-to-SMA bulkhead pigtails; final cable length remains a routed sample gate.

6. MODEM CONTROL CONTRACT

Control outputs originate at the system GPIO expander. FULL_CARD_POWER_OFF and RESET require final modem-specific timing validation.

TP7201–TP7208 are copper factory probes. USB 2/3 has no branch or generic test header.

RF coax leaves the module directly for four right-side bulkhead antennas
FSA2567 defaults to physical SIM 1; open-drain low selects physical SIM 2
Dedicated 3.8 V / 6 A rail; verify selected modem peak-current waveform
M.2 B-key modem socket supports USB 3.0 and USB 2.0 fallback

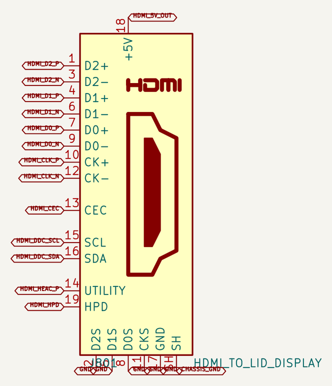
ProComm

Sheet: /WWAN / Dual SIM/
File: WWAN-SIM.kicad_sch

Title: Radxa CM5 ProComm Carrier – WWAN and Dual SIM

Size: A1 | Date: 2026-08-14 | **Rev: A1**
KiCad E.D.A. 10.0.5-rc1 | Id: 4/8

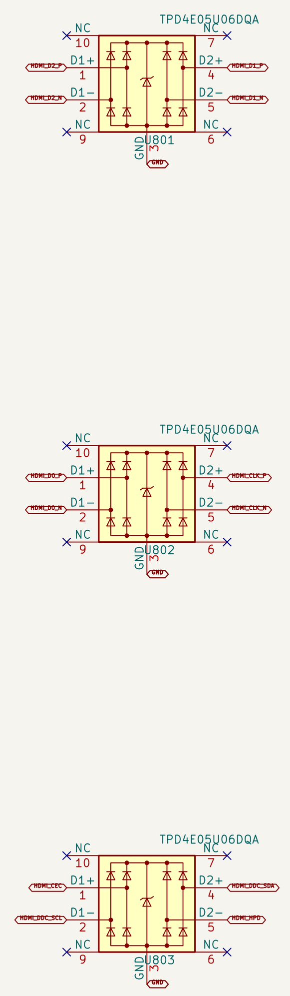
1. HDMI TYPE-A SOURCE CONNECTOR



HDMI_HEAC_N remains unused in Rev A; the selected monitor uses video/DDC/CEC/HPD only.

Shell bonds to chassis at the connector. TMDS shield returns use the local digital ground plane.

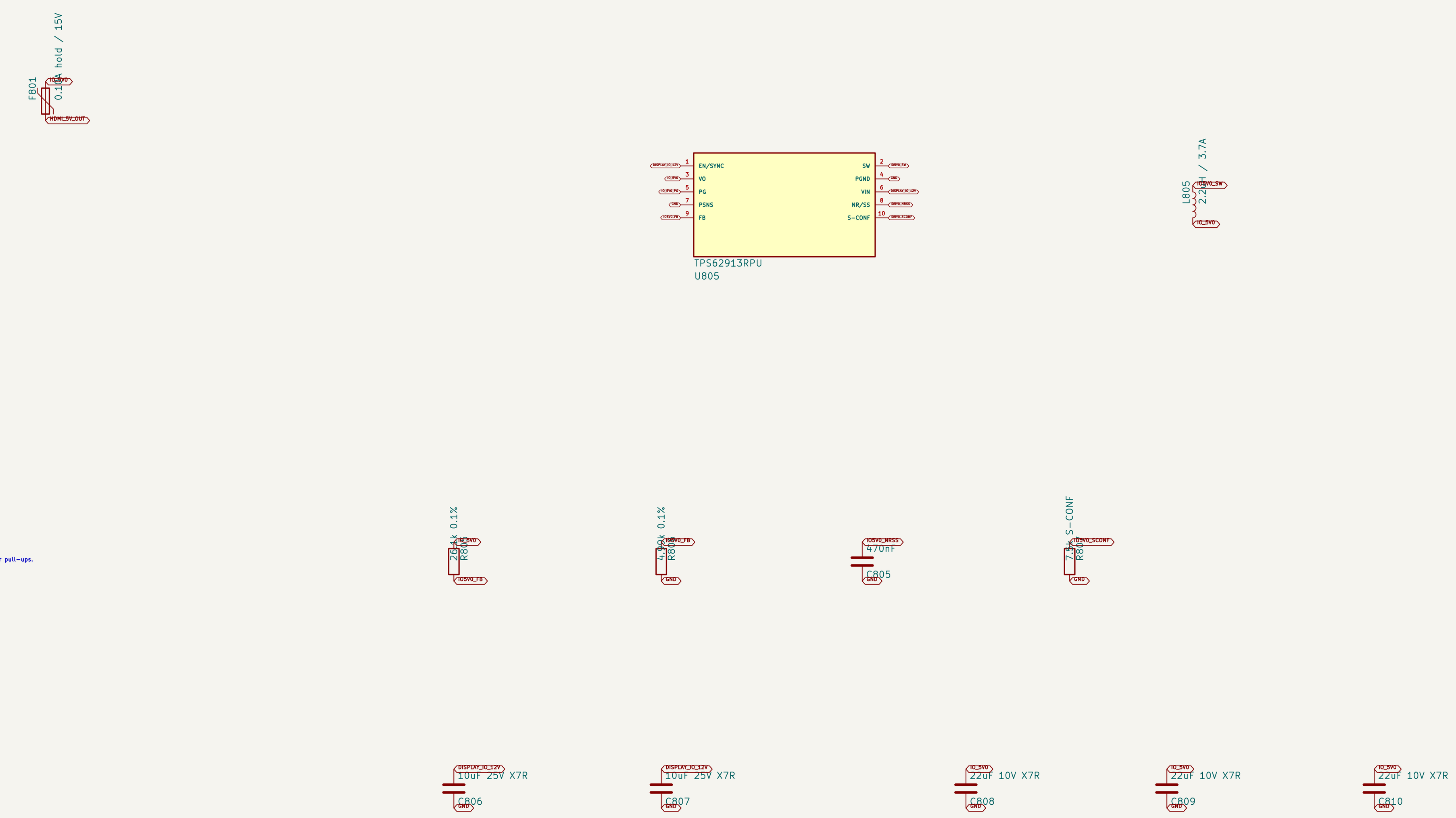
2. HDMI ESD / LOW-CAPACITANCE SHUNTS



Place all HDMI ESD arrays immediately behind J801 with no stubs on TMDS pairs.

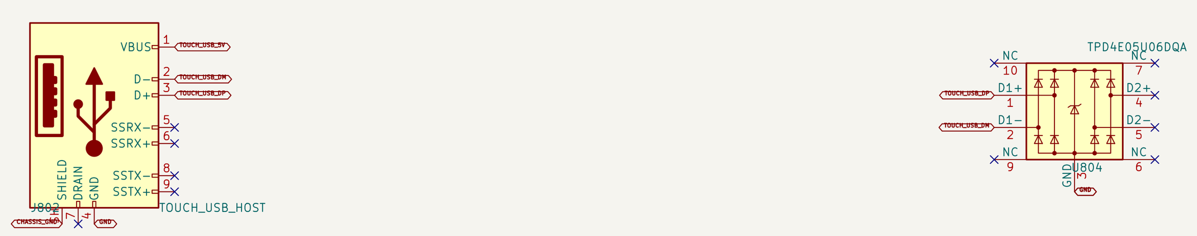
HDMI DDC connects directly to the CM5 5 V DDC pins, matching Radxa CM5 IO V2.2; do not add duplicate carrier pull-ups.

3. DEDICATED IO_5V0 / 2 A BUCK



IO_5V0 feeds CM5 U13-B pin 106, HDMI source 5 V and USB-touch VBUS; no monitor-power eFuse.

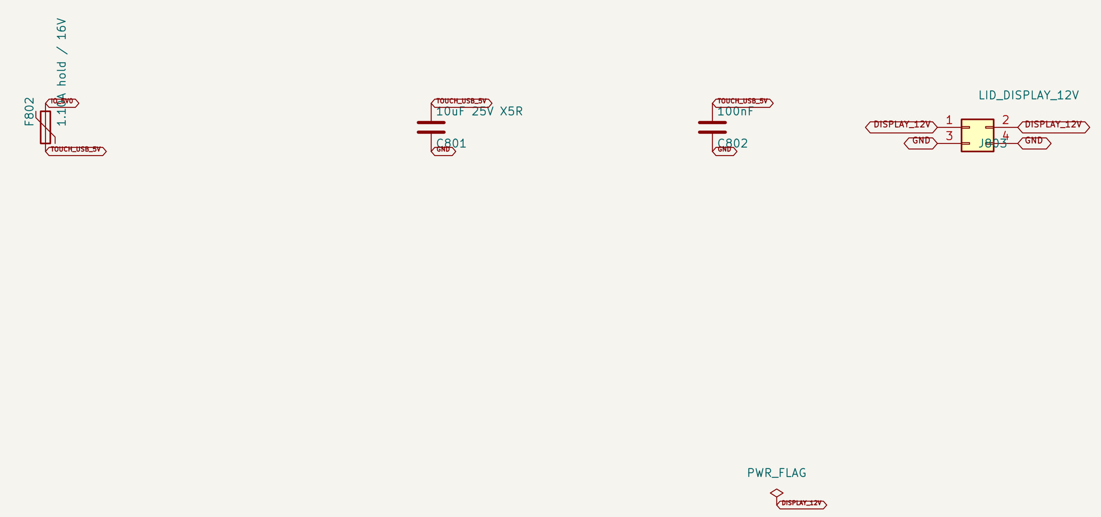
4. USB TOUCH HOST PORT



Only USB 2 is allocated from the CM5. A standard USB 3 A-to-B cable still carries USB 2 touch data.

The monitor-side SuperSpeed Type-B receptacle is not duplicated as a device connector on this host carrier.

5. 12 V / 2.5 A DISPLAY POWER



DISPLAY_12V design load: 2.5 A continuous / 30 W branch, covering the locked 25 W monitor.

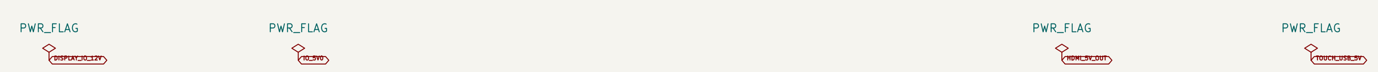
No local display eFuse. Use the upstream AUX_12V branch fuse and H03C keyed 18 AWG harness.

6. HINGE-EDGE HARNESS BUILD

H03A HDMI, H03B USB touch, H03C 12 V: 1000 +/− 25 mm each, with separate retained hinge and service loops.

Acceptance: full lid travel plus 300 mm panel lift / 45 degree tilt; no connector tension or hinge-line bend.

Carrier connectors face down. Store released loops clear of fans, PSU airflow, board keepouts and sharp edges.



No display eFuse; separate fused DISPLAY_12V and DISPLAY_IO_12V branches feed this sheet
USB-A is the CM5 host end; cable terminates in USB-B SuperSpeed at the monitor
Carrier connectors face downward at the hinge-edge harness notch
15.6-inch lid monitor: HDMI video, USB 2 touch, 12 V / 2.5 A

ProComm

Sheet: /Display Harness/
File: Display-Harness.kicad_sch

Title: Radxa CM5 ProComm Carrier – Lid Display Harness

Size: A1	Date: 2026-08-14	**Rev: A1**
KiCad E.D.A. 10.0.5-rc1		Id: 5/8

1. I2S0/TDM PROGRAM-AUDIO LVDS LINK

J901 mates with AUDIO-8X8 J101. Route five 100 ohm differential pairs with the assigned interleaved returns.

SN65LVDT2D includes the ADC-data receiver termination; do not add a duplicate far-end resistor.

2. I2S1 HEADSET LEVEL TRANSLATION

PCA9306 VREF2 and EN share HS_I2C_BIAS through the required 200 k pull-up to 3.3 V.

CM5 audio GPIO is 3.3 V. ES8316 digital pins remain entirely in the translated 1.8 V domain.

3. DEDICATED LOW-NOISE HEADSET RAILS

R900 is the only headset analog-ground bond. Do not create a second PCB or chassis bond.

4. ES8316 I2S1 HEADSET CODEC

CE is pulled to 1.8 V; all ES8316 digital pins are translated. Decoupling follows the Radxa reference topology.

5. TPA6132A2 HEADPHONE AMPLIFIER

G0=1 and G1=0 select 0 dB gain. EN remains software-controlled for pop-free mute.

6. CTIA HEADSET JACK, MIC BIAS, DETECT, AND ESD

CTIA contact map: tip=L, ring1=R, ring2=GND, sleeve=MIC. Verify isolated switch polarity on a Kycon sample.

J910 is routable for prototype capture, but its drawing-derived footprint is blocked from production until sample/coupon sign-off.

7. HEADSET CONTROL POLICY

HEADSET_POWER_EN and HEADSET_AMP_EN default low so the codec and amplifier remain muted during boot.

Sequence: enable rails, configure ES8316, enable TPA6132A2, then ramp software volume.

Firmware must debounce HS_JACK_DET_N and perform pop-free mute before any route change.

Kycon STX-353K7A is CTIA-mapped; its preliminary land pattern requires a coupon
TPA6132A2 provides 0 dB dedicated stereo headphone amplification
I2S1 uses explicit 3.3 V to 1.8 V translation before the ES8316 codec
I2S0/TDM program audio is buffered over LVDS to the separate AUDIO-8X8 board
ProComm

Sheet: /Audio Control / Headset/
File: Audio-Control.kicad_sch

Title: Radxa CM5 ProComm Carrier — TDM and Headset Audio

Size: A0 | Date: 2026-08-16 | **Rev: A1**
KiCad E.D.A. 10.0.5-rc1 | Id: 6/8

Component values are Rev A1 starting values; bench loop, thermal, and transfer validation remain ma
DISPLAY_12V is a fused 12 V / 2.5 A harness branch with no dedicated eFuse
The 27.2 mF / 50 V no-blink bank resides on floor-mounted PWR-SELECT
RAW_OUT_LOAD accepts 10.5 V to 30 V after source selection and load telemetry
ProComm
Sheet: /Power Regulators/
File: Power-Regulators-A1.kicad_sch
Title: Radxa CM5 ProComm Carrier — Power Regulators and Sequencing

Size: A0	Date: 2026-08-14	**Rev: A1**
KiCad E.D.A. 10.0.5-rc1		Id: 7/8


1. 1.8 V TO 3.3 V SYSTEM I2C CONTROL BUS
2. THREE 16-BIT GPIO EXPANDERS
3. EMC2305 FOUR-FAN CONTROL WITH FAILSAFE
4. THREE HIGH-ACCURACY TEMPERATURE SENSORS
5. SIX 12 V TOP-PANEL STATUS-LAMP DRIVERS
6. INDEPENDENT WARM-WHITE PANEL LIGHTS
2N7002K-7
Q1065
TCA9535 devices at 0x20/0x21/0x22 capture status and drive safe control outputs
TMP117 sensors monitor CM5, modem, and board/power zones at 0x48/0x49/0x4A
PWM pull-ups provide full-speed hardware fallback if the controller or firmware fails
EMC2305 controls CPU, modem, intake, and exhaust fans independently
ProComm
Sheet: /Thermal / Fans / IO/
File: Thermal-IO.kicad_sch
Title: Radxa CM5 ProComm Carrier — Thermal and GPIO Control
Size: A0
Date: 2026-08-14
Rev: A1
KiCad E.D.A. 10.0.5-rc1
Id: 8/8